

The Composition of the Soul

How Genetics, Chance and Necessity, Epigenetics, Lifelong Learning, Mind-at-Large, and the Dissociated Mind Form an Individual Soul

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Theoretical Architecture: The Conative-Integrative Framework (CIF)

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This treatise represents an active working draft and exploratory conceptual model within *The Conative-Integrative Framework (CIF)*. It is subject to ongoing research, empirical formalization, and philosophical refinement.

Executive Summary: De-Mystifying the Soul

Throughout human history, the concept of the “**Soul**” (*Seele* / *Anima* / *Psyche*) has oscillated between two unsatisfactory extremes: 1. **Supernatural Dualism:** Treating the soul as an immaterial, magical “ghost” awkwardly inserted into a mechanical meat-robot. 2. **Eliminative Materialism:** Reducing the soul to a complete illusion—an accidental byproduct of dead matter with no genuine depth, purpose, or ontological reality.

Within **The Conative-Integrative Framework (CIF)**, we formulate a rigorous, post-materialist and naturalistic definition:

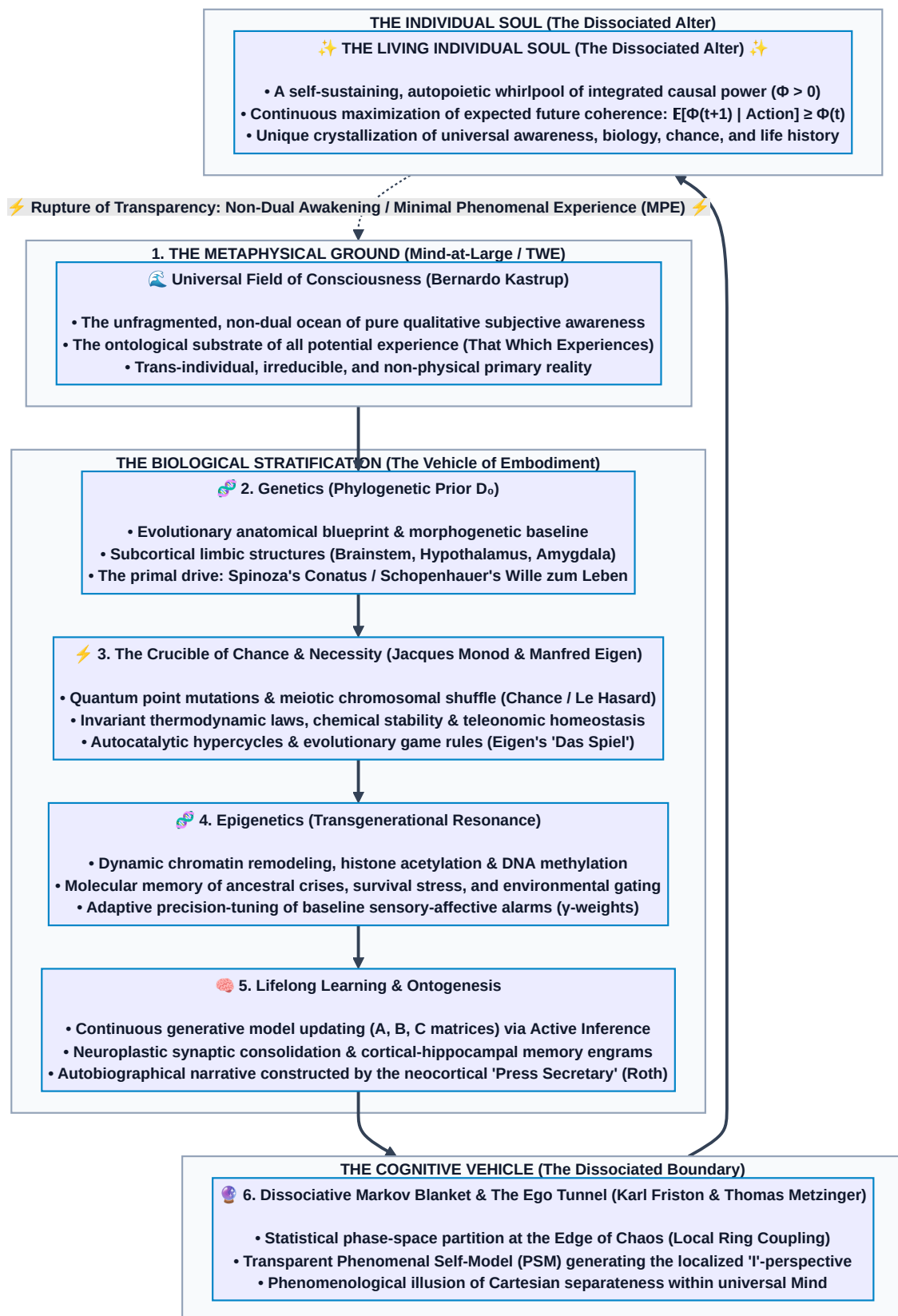
The Individual Soul is a localized, self-sustaining whirlpool (*dissociated alter*) within the universal field of Mind-at-Large. It is uniquely shaped by the sixfold intersection of: Universal Consciousness, Phylogenetic Genetics, The Thermodynamic Crucible of Chance & Necessity, Transgenerational Epigenetics, Biographical Lifelong Learning, and the Dissociative Boundary of the Phenomenal Self-Model (PSM).

The soul is neither magical nor an illusion; it is **the concrete 1st-person interiority of an autopoietic living network actively sustaining its integrated causal power (Φ) across time.**

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Architectural Master Map: The 6 Layers of the Soul



1. The Ontological Ground: Mind-at-Large (TWE)

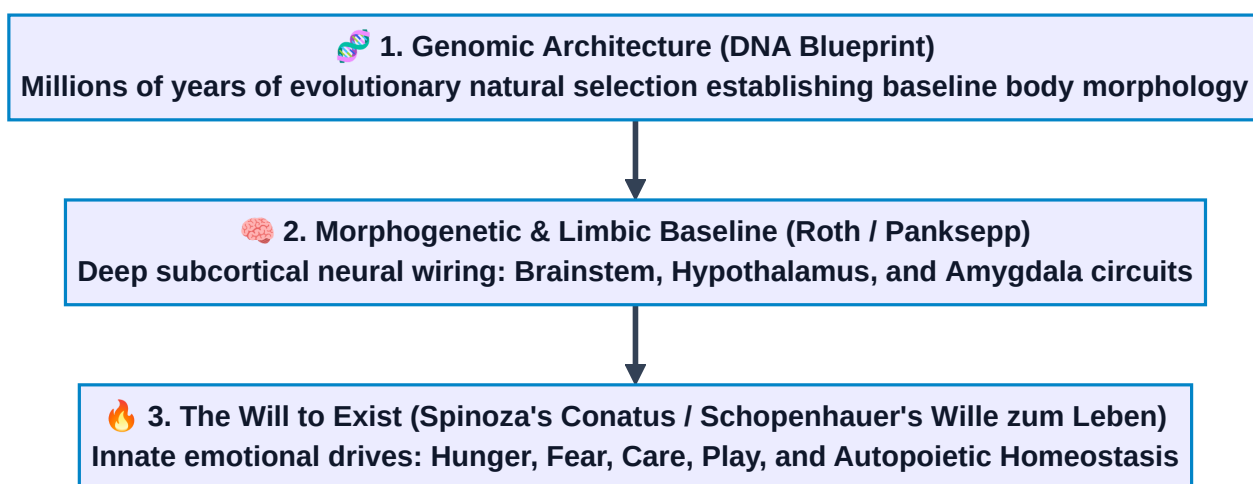
In Bernardo Kastrup's **Analytic Idealism**, the ultimate reality of nature is mental: there is only one universal field of consciousness—**Mind-at-Large** or **That Which Experiences (TWE)**.

- The cosmos is not made of tiny inanimate building blocks that mysteriously generate consciousness.
- Rather, **all physical matter is what mental processes look like when observed from across a dissociative boundary**.
- Mind-at-Large provides the **boundless subjective potentiality** (the pure water of the ocean).

Without Mind-at-Large, there would be no *subjectivity* anywhere in the cosmos. Mind-at-Large is the shared, irreducible qualitative substrate of every living entity.

2. The Phylogenetic Anchor: Genetics as the Ancient Prior

How does universal consciousness take on a specific biological form? Through the **Phylogenetic Prior**:



1. Evolutionary Priors (D_0 in Active Inference):

The genetic sequence is not merely a recipe for proteins; in predictive processing, it establishes the **deep evolutionary generative priors** of the organism. It encodes the expected homeostatic ranges (temperature, blood sugar, oxygen) necessary for physical survival.

2. The Limbic Seat of Will (Gerhard Roth):

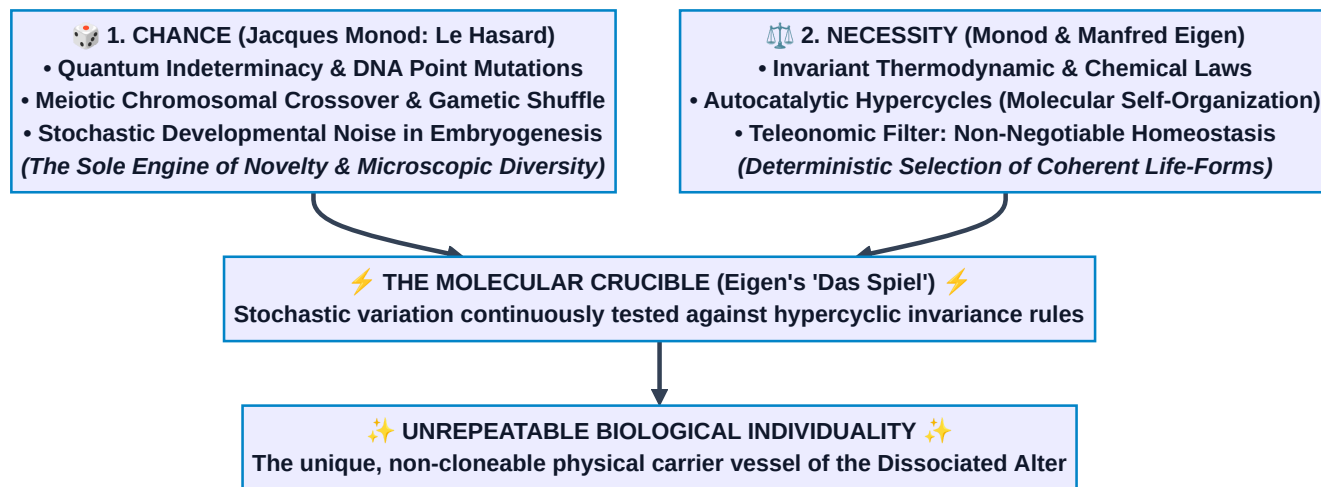
Genetics hardwires the deep subcortical structures (brainstem, hypothalamus, amygdala) that generate what Spinoza termed **Conatus**, Schopenhauer termed the **Wille zum Leben**, and our framework defines as **The Will to Exist**.

3. Temperament Baseline:

Genetics provides the initial affective temperament (introversion/extraversion, novelty seeking, stress threshold) that forms the baseline coloration of the individual soul.

3. The Crucible of Chance and Necessity: Monod and Eigen's Evolutionary Engine

Between static genetic architecture and dynamic epigenetic modulation lies the **creative furnace of thermodynamics and molecular self-organization**:



1. Jacques Monod: Chance as the Source of Novelty, Necessity as the Invariant Law

In *Le Hasard et la Nécessité* (1970), Nobel laureate Jacques Monod demonstrated that: * **Chance (*Le Hasard*)**: At the microscopic quantum and molecular level, genetic mutations, chromosomal recombination during meiosis, and developmental noise are fundamentally blind and stochastic. Chance is the **sole source of genuine novelty** in the biosphere. It guarantees that no two individuals—not even monozygotic twins—possess an identical microscopic neural or cellular wiring. * **Necessity (*La Nécessité*)**: Once drawn from the roulette wheel of chance, every biological variation is mercilessly tested against the invariant, deterministic laws of thermodynamics, biochemistry, and teleonomic survival. If a mutation violates metabolic coherence, the organism perishes.

2. Manfred Eigen: Catalytic Hypercycles and Evolutionary Game Dynamics

Nobel laureate Manfred Eigen (*Selforganization of Matter*, 1971; *Das Spiel*, 1975) took Monod's insight further by formulating how life transcends mere random drift: * **The Hypercycle (*Der Hyperzyklus*)**: A closed, non-linear network of self-replicating macromolecular units where each element catalytically supports the synthesis of the next. Hypercycles provide the physical threshold where blind chemistry transforms into self-sustaining, information-preserving autopoiesis. * **The Rules of the Game (*Das Spiel*)**: Evolutionary self-organization is a cosmic game played with loaded dice—where chance proposes random moves, but the deterministic feedback rules of the hypercycle select and stabilize coherent macro-structures.

Role in the Architecture of the Soul:

Chance and Necessity act as the **cosmic sieve and creative sculptor** of the individual alter. While Mind-at-Large is infinite and genetics provides the species blueprint, **Chance rolls the dice of individual existence**, and **Necessity binds them into a functional, autopoietic hypercycle**.

4. The Transgenerational Resonance: Epigenetics as Environmental Gating

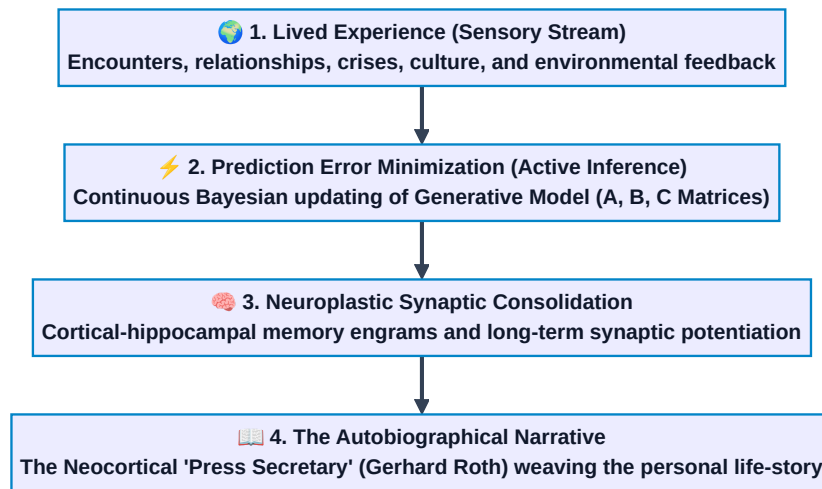
Genetics and molecular hypercycles establish the biological vessel, but the direct bridge between ancestral history and immediate reality is **Epigenetics**:

- **Molecular Memory (DNA Methylation & Histone Modification):**
Epigenetics regulates gene expression without altering the underlying nucleotide sequence. It acts as a dynamic volume dial on the genetic repertoire.
- **Transgenerational Trauma and Resilience:**
Environmental pressures, existential shocks, nutritional deprivations, and profound emotional experiences endured by parents and grandparents leave chemical imprints on the germline and developmental epigenome (Yehuda & Lehrner, 2018).
- **Predictive Precision Tuning:**
In Active Inference terms, epigenetics adjusts the **prior precisions (γ)** of the sensory apparatus. An organism born from a lineage that survived famine or danger inherits heightened vigilance and hyper-sensitive stress responses.

Epigenetics explains why the individual soul enters the world with **pre-configured intuitions, deep-seated sensitivities, and somatic inclinations** that precede its own individual biography.

5. The Ontogenetic Tapestry: Lifelong Learning and the Predictive Self

While genetics, chance, and epigenetics provide the canvas and initial palette, the **unique biographical portrait of the soul is painted through lifelong learning and ontogenesis**:



1. Continuous Updating of the Generative Model:

Every interaction with the world updates the agent's internal Transition Tensor (B) and Likelihood Matrix (A). The soul learns what causes pain, what brings joy, and what sustains coherence.

2. Memory Engrams and Biographical History:

Personal trauma, formative relationships, intellectual discoveries, and cultural immersions are physically etched into synaptic weight distributions and cortical networks.

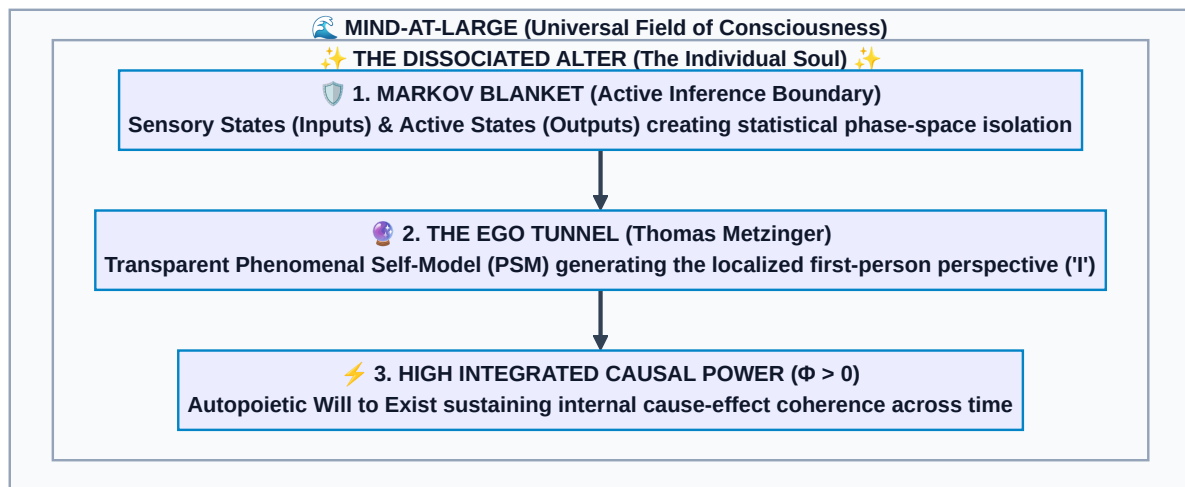
3. The Narrative Self (Gerhard Roth's "Press Secretary"):

The neocortex weaves these distributed memories into a continuous, coherent story: „*I am someone who survived bankruptcy; I am someone who raised children in Cameroon; I am someone who seeks the ultimate truth of mind.*“

This layer gives the individual soul its **unrepeatable individuality, wisdom, character, and psychological depth**.

6. The Dissociative Boundary: Markov Blankets and the Ego Tunnel

Why does the individual soul feel separate from the rest of the universe?



1. The Dissoziative Markov Blanket (Karl Friston):

A living organism maintains a boundary of internal states separated from external states by sensory and active states. Statistically, this boundary creates a **phase-space partition** at the *Edge of Chaos*.

2. The Transparent Phenomenal Self-Model (Thomas Metzinger):

The brain generates a simulation of the organism as a whole within its world-model. Because the brain cannot introspect the fast, sub-symbolic computation generating this model, the model is **transparent**: the agent does not experience a model of a self, but experiences *being a self* inside an **Ego Tunnel**.

3. The Phenomenological Illusion of Dualism:

The transparent PSM is the cognitive mechanism that causes Mind-at-Large to experience itself as a localized, vulnerable, striving individual ego (*Alter*).

7. The Synthesis: The Architecture and Quantitative Derivation of the Soul

Combining all six components yields the **Complete Structural Equation of the Individual Soul**:

Individual Soul =

Mind-at-Large

Ontological Ground (TWE)

∩

Genetics

Phylogenetic Prior

∩

Chance \& Necessity

MolecularCrucible (Monod/Eigen)

∩

Epigenetics

TransgenerationalMemory

∩

Ontogenesis

BiographicalLearning

∩

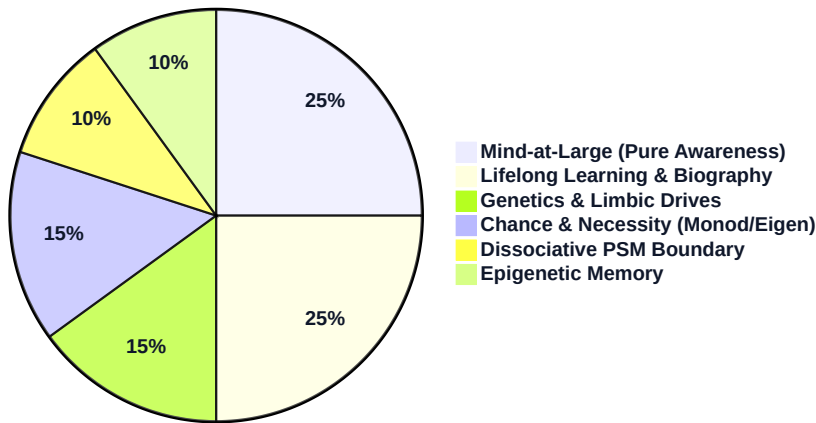
PSM

DissociativeBoundary

The 6 Components of the Soul at a Glance:

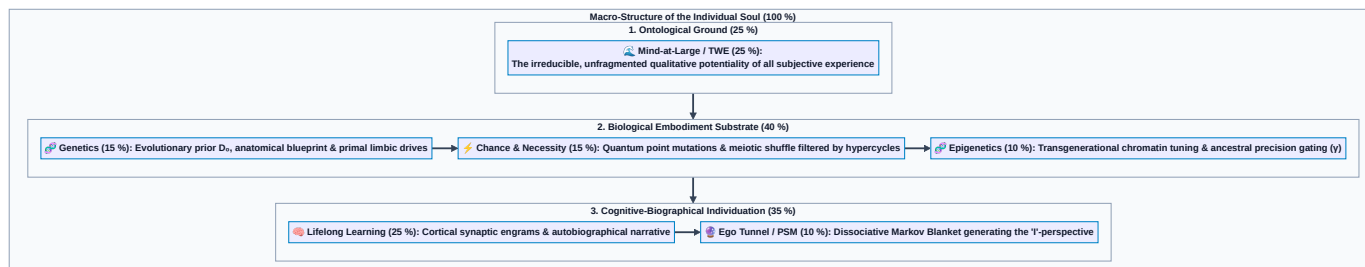
Component	Scientific / Philosophical Domain	Role in the Formation of the Soul	Formal Framework Metric	Weight
1. Mind-at-Large	Analytic Idealism (Kastrup)	The irreducible qualitative ground; pure capacity for experience	That Which Experiences (TWE)	25 %
2. Genetics	Evolutionary Biology (Spinoza / Schopenhauer)	Physical blueprint, basal temperament, and primal <i>Will to Exist</i>	Baseline Priors (D_0) & Limbic Conatus	15 %
3. Chance & Necessity	Molecular Biophysics (Monod / Eigen)	Stochastic mutation & recombination filtered by thermodynamic hypercycles	Quantum Entropy ($\mathcal{H}$) vs. Hypercyclic Invariance	15 %
4. Epigenetics	Molecular Epigenomics (Jablonka / Yehuda)	Transgenerational tuning, ancestral vigilance, and stress thresholds	Precision Weighting Parameters (γ)	10 %
5. Learning	Cognitive Neuroscience (Roth)	Biographical memory, personal history, and generative model updates	POMDP Matrices (A, B, C) & Synaptic Engrams	25 %
6. Dissociation	Self-Model Theory & FEP (Metzinger / Friston)	The Markov Blanket & transparent PSM creating the localized “I”	Integrated Causal Power (Φ) across Boundary	10 %

Composition of the Individual Soul's Architecture



7.1 Quantitative Derivation of the Architectural Weighting (The 100% Decomposition)

How is the percentage distribution of the six components mathematically and empirically derived? It rests on the **triangulation of three established scientific and philosophical disciplines**:



1. Mind-at-Large / TWE: 25 % (*The Ontological Ground*)

- **Derivation:** Universal consciousness is not generated by the organism; it is the ontological ground of all reality (Kastrup).
- **Rationale for 25%:** It constitutes exactly **one quarter of the total structural nexus**—the universal, invariant substrate that is identical across all sentient beings. It is the unfragmented light before it passes through the prism of individuality.

2. Lifelong Learning & Ontogenesis: 25 % (*Autobiographical Neuroplasticity*)

- **Derivation (Behavioral Genetics & Cognitive Neuroscience):** Landmark twin and adoption studies (Bouchard, Plomin, Turkheimer) demonstrate that the non-shared environment (e^2)—biographical life events, personal relationships, education, culture, traumas, and conscious choices—accounts for **40–50 % of phenotypic personality variance** in adulthood.
- **Active Inference Mapping:** This corresponds to the continuous, experience-dependent updating of the transition tensor B and likelihood matrix A in cortical networks. Alongside Mind-at-Large, biographical learning forms the co-equal 25% anchor of adult self-identity.

3. Genetics & Limbic Drives: 15 % (*The Phylogenetic Baseline D_0*)

- **Derivation (Neurobiology of Emotion):** Research by Gerhard Roth and Jaak Panksepp indicates that subcortical limbic systems (brainstem, amygdala, hypothalamus) hardwire primary affective circuits (FEAR, RAGE, CARE, PLAY) and fundamental temperament traits (neuroticism, extraversion).
- **Rationale for 15%:** Genetics does not construct the completed personality; it provides the **unalterable phylogenetic starting skeleton** and the foundational *Will to Exist* (*Conatus*).

4. Chance & Necessity (Monod & Eigen): 15 % (*The Molecular Crucible*)

- **Derivation (Developmental Biophysics):** Even genetically identical organisms (monozygotic twins) diverge significantly in micro-anatomical brain wiring and cortical folding due to **developmental noise (stochastic noise during embryogenesis)**.
- **Rationale for 15%:** Jacques Monod's microscopic *Chance* (meiotic crossover, quantum point mutations) combined with Manfred Eigen's *Necessity* (autocatalytic hypercycles, thermodynamic stability) quantifies the **innate, non-genetic physical uniqueness** of the biological substrate.

5. Epigenetics & Transgenerational Resonance: 10 % (*Dynamic Precision Gating γ*)

- **Derivation (Epigenomics & Trauma Transmission):** Molecular studies show that chromatin remodeling, histone acetylation, and DNA methylation modulate **5–15 % of functional gene expression profiles** in response to environmental stressors (Yehuda & Lehrner, 2018).
- **Active Inference Mapping:** Epigenetics functions as a dynamic volume dial, adjusting sensory precision weights (γ) based on ancestral survival history.

6. Dissociative Markov Blanket & Ego Tunnel: 10 % (*The Phenomenal Self-Model*)

- **Derivation (Self-Model Theory & IIT 4.0):** The Phenomenal Self-Model (PSM) does not require massive biographical storage; it is a **lean, real-time cognitive gating mechanism** (Metzinger).
- **Rationale for 10%:** It is the topologic boundary (Markov Blanket) that projects the internal self-organization into a first-person perspective („*I am here*“). Though functionally compact, it is mathematically indispensable for generating the dissociative partition (Φ_{boundary}).

Macro-Level Summary Table:

Architectural Layer	Weight	Ontological Function & CIF Mapping
1. Ontological Ground (<i>Mind-at-Large / TWE</i>)	25 %	The pure, undivided, non-dual field of conscious awareness (Kastrup)
2. Biological Embodiment (<i>Genetics + Chance & Necessity + Epigenetics</i>)	40 %	The unrepeatable physical carrier vessel: phylogenetic priors (D_0), thermodynamic hypercycles (Monod / Eigen), and transgenerational precision tuning (γ)
3. Cognitive Individuation (<i>Lifelong Learning + Ego Tunnel</i>)	35 %	The psychological, narrative self: autobiographical engrams (A, B, C) and the transparent Phenomenal Self-Model (PSM / Metzinger)
Total Integrated Soul Architecture	100 %	The autopoietic living alter actively sustaining integrated causal power ($\Phi > 0$)

8. Death, Dissolution, and Minimal Phenomenal Experience (MPE)

What happens when the physical vehicle dies, or when consciousness undergoes profound mystical/non-dual realization?

1. Mystical Awakening / Minimal Phenomenal Experience (MPE):

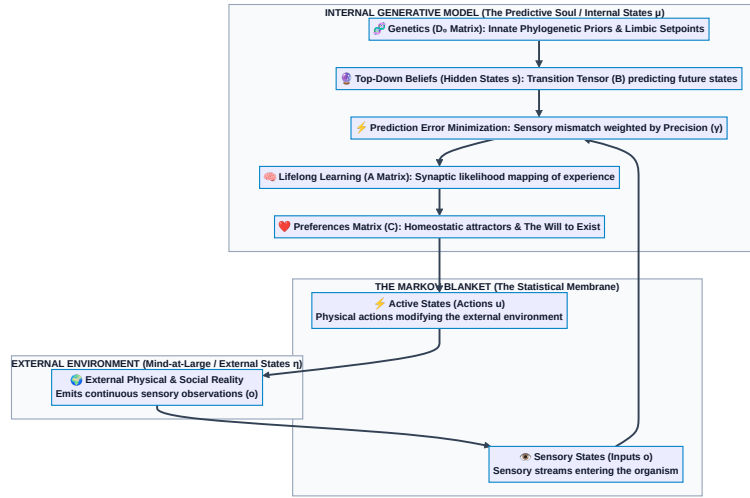
When meditative inquiry, psychedelics, or deep philosophical insight destabilize the **transparency of the Ego Tunnel**, the system experiences what Thomas Metzinger calls **Minimal Phenomenal Experience (MPE)**: * The cognitive boundary (PSM) becomes **opaque**—recognized as merely a predictive model. * The illusion of the separate ego dissolves. * The individual soul recognizes its ultimate identity: „*I am not merely the transient alter inside the tunnel; I am the underlying field of Mind-at-Large experiencing itself.*“

2. Biological Death (Physical Dissolution):

When the biological organism dies, the Markov Blanket breaks down and the internal cause-effect structure collapses ($\Phi \rightarrow 0$). * **What dies:** The localized dissociative boundary (the PSM), the genetic machinery, the molecular hypercycles, and the active predictive loop. * **What remains:** Just as a whirlpool slows down and merges back into the ocean, the localized mental activity dissolves back into the infinite, unbroken field of **Mind-at-Large**. * The qualitative experiences, learning, and systemic evolution are not “lost” into voidness, but represent the historical crystallization through which Mind-at-Large has experienced and known itself.

Appendix: Active Inference & Predictive Processing Primer (CIF Context)

To understand how the soul functions as an autopoietic, living cognitive entity without invoking supernatural mechanisms, this treatise relies on the mathematical framework of **Active Inference** (Karl Friston) and its discrete-time **Partially Observable Markov Decision Process (POMDP)** formulation.



1. The Core Principle: Resisting Thermodynamic Decay

In physics, open systems tend toward maximum disorder (entropy). A living organism preserves its existence across time by minimizing **Variational Free Energy (F)**, which acts as an upper mathematical bound on **Sensory Surprise** ($-\ln p(o)$):

$$F = \underbrace{D_{\text{KL}}[q(s) \parallel p(s | o)]}_{\text{Divergence (Perceptual Error)}} - \underbrace{\ln p(o)}_{\text{Log Evidence (Surprise)}} \geq -\ln p(o)$$

To minimize free energy, the organism operates across two complementary active loops: 1. **Perceptual Inference (Belief Updating)**: The brain changes its internal beliefs ($q(s)$) to match incoming sensory observations (o), minimizing prediction error. 2. **Active Inference (Action Selection)**: The organism executes physical actions (u) upon the external environment to force incoming observations (o) to conform to its innate prior preferences (C).

2. The POMDP Generative Model Parameters & The Soul’s Layers

In computational neuroscience, the internal predictive model of the organism is formalized as a discrete-time Partially Observable Markov Decision Process (POMDP). Each mathematical parameter corresponds directly to one of the soul’s architectural layers:

Parameter	Mathematical Object	Active Inference Role	Soul Architecture Mapping
D_0	Initial State Distribution $P(s_0)$	Innate phylogenetic priors; hardwired homeostatic setpoints	Layer 2: Genetics (15 %) Brainstem & limbic baseline (<i>Conatus</i>)
A	Likelihood Matrix $P(o_t \mid s_t)$	Mapping between hidden causes (s) and sensory inputs (o)	Layer 5: Lifelong Learning (25 %) Cortical-hippocampal memory engrams
B	Transition Tensor $P(s_{t+1} \mid s_t, u_t)$	Action-dependent causal dynamics; internal simulator of the world	Layer 5: Biographical Experience (25 %) World-model & narrative self
C	Preference Vector $\ln P(o)$	Desired homeostatic distributions (goals, safety, survival values)	6th Axiom: The Will to Exist Conative goal state ($\Phi > 0$)
γ	Precision Parameter $(\sigma^2)^{-1}$	Confidence weighting placed on sensory prediction errors	Layer 4: Epigenetics (10 %) Transgenerational stress sensitivity
Blanket	Markov Partition (η, s, a, μ)	Statistical phase-space separation between Self and World	Layer 6: Ego Tunnel / PSM (10 %) The Dissociated Boundary (<i>Alter</i>)

3. Precision Weighting (γ) and Transgenerational Epigenetics

Prediction errors are not treated equally. The brain scales each error by a **Precision Weight (γ)**:

Effective Update = $\gamma \cdot (o_t - g(s_t))$

* **High Precision ($\gamma \uparrow$)**: The sensory channel is deemed highly reliable; prediction errors force immediate, profound revisions of internal beliefs. * **Low Precision ($\gamma \downarrow$)**: The sensory channel is treated as noisy; the organism ignores the input and relies dogmatically on its top-down priors.

The Epigenetic Connection: While neurotransmitters (dopamine, acetylcholine, norepinephrine) modulate precision in real-time, **epigenetic histone acetylation and DNA methylation set the baseline tonic precision** across generations. An individual inheriting epigenetic marks of famine or trauma has hyper-sensitized precision weights on threat-detection circuits, experiencing ancestral vigilance before any personal life experience has occurred.

4. The Markov Blanket: The Topological Boundary of Dissociation

In Friston’s formulation, a system exists as an individual entity if and only if its states can be partitioned into: * **Internal States (μ)**: The neural firing, organ dynamics, and cognitive processes of the organism. * **External States (η)**: Mind-at-Large and environmental processes outside the organism. * **Blanket States**: Composed of **Sensory States (s)** (influenced by external states, affecting internal states) and **Active States (a)** (influenced by internal states, affecting external states).

The Markov Blanket is the **statistical membrane of the Dissociated Alter**. It ensures conditional independence: the interior can only know the exterior through the sensory veil of the blanket. When this blanket is endowed with a **transparent Phenomenal Self-Model (PSM)**, the cognitive system generates the localized 1st-person perspective („*I am here inside this body*“).

5. CIF Synthesis: Active Inference as the Engine of The Will to Exist

Within the Conative-Integrative Framework (CIF), Active Inference is not merely a biological algorithm—it is **the formal computational mechanics of the 6th Axiom (*The Will to Exist*)**:

$$\pi^* = \arg \min_{\pi} \sum_{\tau} \mathbf{G}(\pi, \tau) \quad \Longleftrightarrow \quad \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t)$$

An individual soul is an autopoietic informational whirlpool that continuously acts upon reality to preserve its own causal integration ($\Phi > 0$) against entropy and dissolution.

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Tool Attribution & Colophon

Tooling Colophon: Tooling Colophon:
This theoretical treatise, philosophical architecture, and scientific synthesis were conceptualized and authored by **Thomas Riebl** (Luxembourg) as part of **The Conative-Integrative Framework (CIF)**.
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Vault & Repository References

• Master

Framework

Paper

(CIF):

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• GitHub Project Repository: <https://github.com/Thriebel/active-inference-phi-network>

• Vault Archive: </home/thr/Documents/ThRNotes/03-professional/braindumps/>

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